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Numerical Methods

Lab 4

Newton Method for $\boldsymbol{f(x) = x\sin(4x)}$

1-Aim of the Lab

In this lab, Newton's root-finding method was applied to the function:

$$f(x) = x\sin(4x)$$

The main aim is to observe how the initial guess affects the root found by the algorithm and the number of iterations required for convergence. Since this function has many roots, Newton's method may converge to different roots depending on the starting point.

The lab also includes the implementation of stopping criteria for the Newton method.

2-Stopping Criteria

In root-finding methods, stopping criteria are used to decide when the algorithm should stop. In this lab, three stopping criteria were considered.

The first stopping criterion is the maximum number of iterations:

$$n \leq n_{max}$$

This prevents the algorithm from running forever.

The second stopping criterion is the accuracy of the approximation. For Newton's method, the difference between two successive approximations is checked:

$$| x_n - x_{n-1} | \leq TOLX$$

If the new approximation is very close to the previous one, the algorithm stops.

The third stopping criterion is based on the function value:

$$| f(x_n) | \leq TOLF$$

If the value of the function is close enough to zero, the algorithm accepts the current approximation as a root.

These stopping criteria are also given in the Lab 4 instruction file.

3-Function and Derivative

The function used in this lab is:

$$f(x) = x\sin(4x)$$

For Newton's method, the derivative of the function is needed. The derivative is:

$$f'(x) = \sin(4x) + 4x\cos(4x)$$

Newton's method uses the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This formula is applied repeatedly until one of the stopping criteria is satisfied.

4-Method

First, Newton's method was tested with one initial guess. In my code, the example initial guess was chosen as:

$$x_0 = 1.0$$

Then, Newton's method was applied for many different initial guesses in the interval:

$$[-5,5]$$

The initial guesses were generated with a step size of 0.05.

For each initial guess, the program recorded:

- the root found by Newton's method,
- the number of iterations,
- whether the method converged or not.

After this, three graphs were plotted:

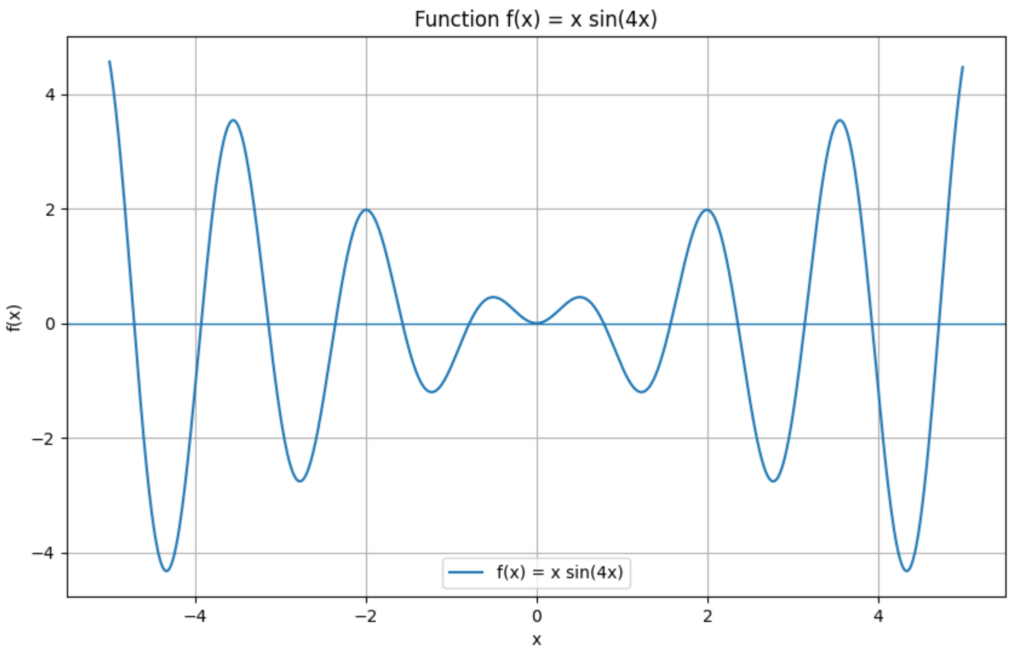
1. Function graph of $f(x) = x\sin(4x)$
2. Initial Guess vs Found Root
3. Initial Guess vs Number of Iterations

The Lab 4 task asks us to run the Newton code in a loop for different initial points and plot how the root and number of iterations depend on the initial guess.

5-Results

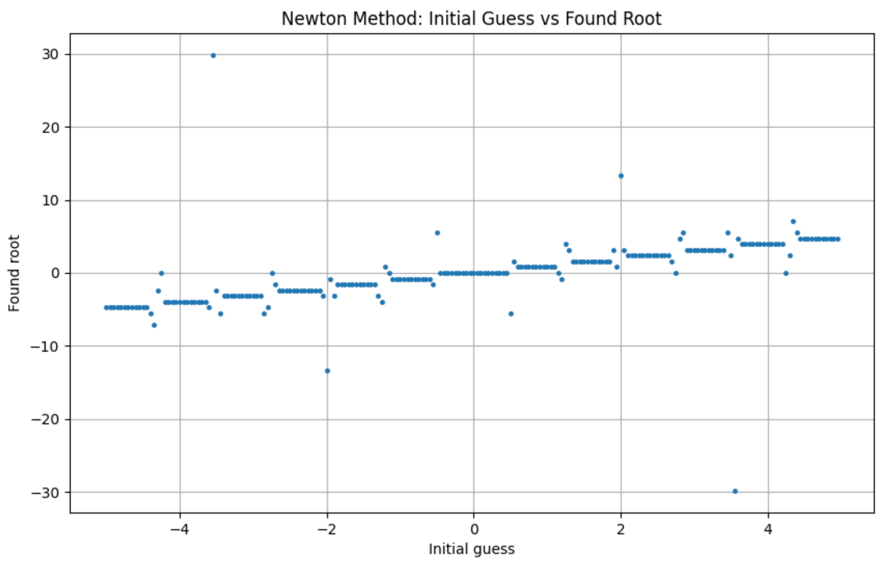
5.1 Function Graph

The graph of the function $f(x) = x\sin(4x)$shows that the function has many roots. These roots occur where the graph crosses the x-axis.



5.2 Initial Guess vs Found Root

The second graph shows the relationship between the initial guess and the root found by Newton's method.

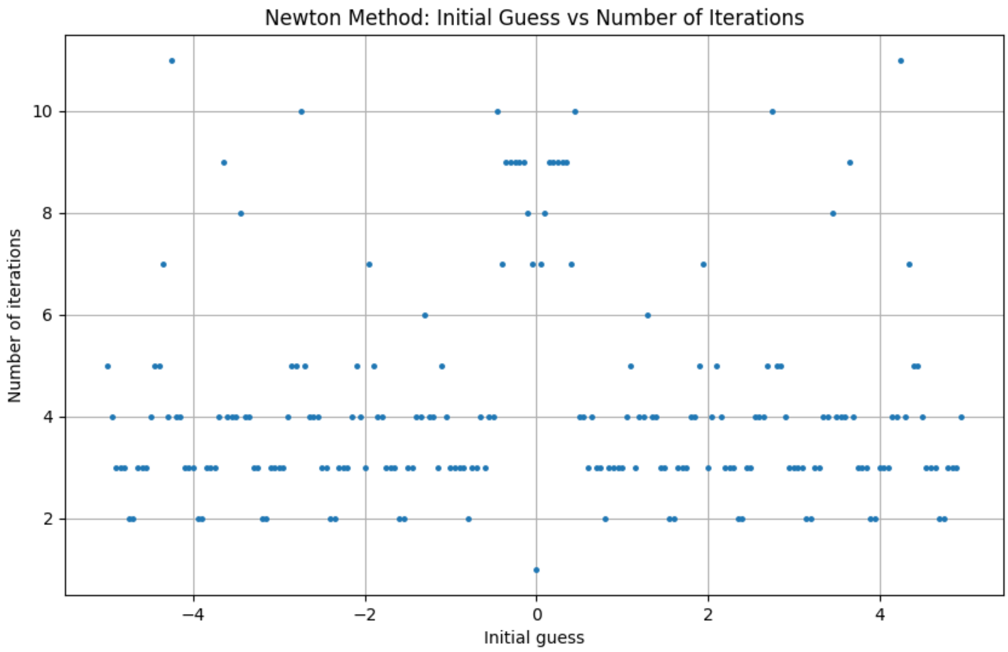


From the graph, it can be seen that the relationship is not smooth. A small change in the initial guess can sometimes cause Newton's method to converge to a different root.

This happens because the function has many roots, and Newton's method is sensitive to the starting point.

5.3 Initial Guess vs Number of Iterations

The third graph shows the number of iterations required for each initial guess.



Most initial guesses converge in a small number of iterations. However, some starting points require more iterations. This means that the speed of Newton's method depends on the chosen initial guess.

6-Discussion

The results show that Newton's method is a fast method for finding roots, but it depends strongly on the initial guess.

For the function:

$$f(x) = x\sin(4x)$$

there are many roots. Because of this, different initial guesses may lead to different roots.

In some cases, the method converges quickly. In other cases, it needs more iterations. This can happen when the derivative is small or when the starting point is in a region where the Newton step makes a large jump.

The scatter plot was used because the found roots do not change continuously with the initial guess. Instead, the method may suddenly jump to another root.

7-Conclusion

In this lab, Newton's method was applied to the function:

$$f(x) = x\sin(4x)$$

The method was tested for many different initial guesses between -5 and 5.

The results show that Newton's method is sensitive to the initial guess. Different starting points may converge to different roots. Also, the number of iterations changes depending on the starting point.

The stopping criteria helped the algorithm stop when the result became accurate enough or when the maximum number of iterations was reached.

Overall, Newton's method is efficient, but choosing a good initial guess is important for reliable and fast convergence.